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SEPTEMBER 25, 1957
SUPERSEDING
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FEDERAL SPECIFICATION

CLEANING COMPOUND, SOAP ABRASIVE, PASTE, FOR PAINTED SURFACES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers a paste cleaner suitable for the removal of grease, dirt, and grime from painted surfaces (woodwork and walls), tile and porcelain, and aluminum utensils and fixtures.

1.2 Classification. — Cleaning compound, soap abrasive, for painted surfaces, shall be of one grade and the following types as specified:

Type I.—Mild or moderate abrasive.

Type II.—Harsh abrasive.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specification:

PPP-C-96—Cans, Metal, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of the Military Standards referenced above, required by contractors in connection with

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specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The product shall consist of a thoroughly blended paste of soap, surfactant, abrasive, humectant, emollient, and water correctant.

3.2 pH.—The pH of the product shall be not less than 9.5 and not more than 10.5 when tested as specified in 4.4.1.

3.3 Rinsing properties.—The product shall rinse satisfactorily when tested as specified in 4.4.2.

3.4 Corrosion.—A ten percent dispersion of the material shall not cause aluminum to discolor, dull or pit when tested as specified in 4.4.3.

3.5 Cleaning efficiency. — The compound shall satisfactorily clean aluminum specimens and porcelain tile squares prepared as specified in 4.4.4.

3.6 Abrasiveness.—With type I cleaner, the gloss reduction of aluminum panels shall not exceed 40 percent; with type II cleaner, it shall not exceed 25 percent when tested as specified in 4.4.5.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all cleaner of the same type manufactured as one batch and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests. When Military Standard MIL–STD–105 specifies an action by the Government, it shall, at the option of the Government, be performed either by the Government or by the contractor under the supervision of the Government Inspector.

4.1.3 *Sampling for tests.*—From each inspection lot, two samples shall be taken as specified in method 101.1 of Federal Test Method Standard No. 536. Samples shall be sealed, marked, and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided that the contractor has removed or repaired all nonconforming containers.

4.3 Lot acceptance tests. — The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If either specimen fails in one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

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4.1 Test procedures.

4.1.1 *pH.*—Dissolve 1.0 grams of cleaner in 100 ml. of freshly boiled and cooled (to room temperature) distilled water, and mix. Determine the pH value of this solution, or a portion of it, to the nearest 0.1 unit at 25° C. by means of a suitable pH meter which employs a glass electrode. No correction for sodium ion shall be made.

4.1.2 *Rinsing properties.*—Swirl 50 ml. of a 10 percent dispersion of cleaner prepared with 7-grain hardness water in a clean 150-ml. Erlenmeyer flask. Pour out the dispersion and allow to drain for 30 seconds. Rinse the flask with 5 separate 50 ml. portions of 7-grain water. Repeat the entire procedure with another sample and flask using distilled water throughout in place of hard water. The flask treated with the hard water shall show no more deposited matter or film than the flask treated with distilled water.

4.1.2.1 The 7-grain hardness water shall be prepared from the following recipe:

Dissolve separately in distilled water 29.6 gm. $MgSO_4.7H_2O$ and 26.4 gm. $CaCl_2.2H_2O$ (ratio of Ca:Mg of 60:40) and make up to 15 liters giving a 2000 p.p.m. water. Dilution to hardness desired:

$$\frac{\text{No. of ml. wanted} \times \text{hardness}}{2000}$$

= (A) ml. of 2000 p.p.m. water to use

Make (A) up to volume with distilled water.

4.1.3 *Corrosion.*—Degrease 0.051 inch x 2½ inches x ⅝ inches strips of aluminum (reflector sheet, standard bright finish in "0" temper[1]) in a mixture of equal parts of alcohol and acetone. Completely dry 2 strips and immerse in 200 ml. of a 10 percent dispersion of cleaner in a 250-ml. beaker. Cover the beaker. Maintain beaker at a temperature of 140° F. for four hours. At end of immersion period remove strips, rinse with distilled water and observe for signs of discoloration, dulling, pitting, etc.

4.1.4 *Cleaning efficiency.* — Mix a fresh batch of soil consisting of equal parts of vegetable shortening, lard and cooking oil. Spread an even coat of this soil on a 0.051-inch x 6 inch square of aluminum (reflector sheet, standard bright finish "0" temper[1]). Bake panel in an electric oven kept at 400° F. for four hours and remove to cool. Prepare a square of porcelain (bathroom) tile by scratching the surface with lead solder, 50/50 bases. Apply one teaspoon (5-8 gm) of cleaner to a cellulose sponge measuring 4 inches x 4 inches. Manually rub the aluminum panel and tile square with the sponge for 20 seconds in a back and forth motion. Then rub for 10 seconds at a right angle to the first rubbing. Observe for removal of soil.

4.1.5 *Abrasiveness.* — Employ a Gardner washability machine[2] equipped with a weighted abrasion boat faced with a piece of felt billiard cloth. Apply 20 ml. of a 1 percent cleaner dispersion to the cloth and polish a 0.051 inch x 6 inch x 6 inch panel of aluminum (reflector sheet, standard bright finish "0" temper[1]) which has been measured for gloss with a Hunter reflectometer at 3 angles of incident light of 45, 60, and 75 degrees. Similarly obtain gloss of panel after polishing for one hundred strokes. Calculate percent loss in gloss averaging the gloss readings (average is meant as the sum of the 45, 60, and 75 degree readings divided by three).

5. PREPARATION FOR DELIVERY

The definitions and applications of the various levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Preservation and packaging.

5.1.1 *Level A.*—The product shall be packaged in 1-pound metal cans conforming to type V, class 2, of Federal Specification PPP–C–96. Plan B exterior coating shall be required.

[1] Alcoa No. 28 obtainable from the Aluminum Company of America and any of its outlets.

[2] Henry A. Gardner Laboratory, Inc., 4723 Elm Street, Bethesda, Maryland.

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5.1.2 *Level C.*—Cleaner shall be packaged according to good commercial practice and permitting delivery in a satisfactory usable condition at the lowest tariff rates.

5.2 Packing.

5.2.1 *Level A.* — Cleaner packaged in 1-pound cans shall be packed in accordance with the appendix of Federal Specification PPP–C–96.

5.2.2 *Level C.*—Cleaner packaged in commercially acceptable containers shall be packed in accordance with commercial practice.

5.3 Marking.

5.3.1 *Civilian agencies.*—Shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking specified in the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Purchasers should exercise any desir options offered herein. (See 1.2, 5.1, and 5.2,.

6.2 Transportation description.—Transportation description applicable to this item is:

Soap, not otherwise indexed by name.
Carload weight—80,000 pounds.
Truckload weight—36,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—QM
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Air Force,

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